RESHAPING SPINE CARE

For decades, spine care relied on methods that now seem outdated. Back then, if you had a slipped disc or severe back pain, the standard advice was bed rest—sometimes for weeks. Today, that's seen as counterproductive: prolonged immobility can weaken muscles, increase stiffness and actually slow recovery. Doctors now nudge patients to get moving sooner, using gentle physiotherapy to rebuild strength and flexibility.

The same rethink is happening on other once-standard treatments. Bulky back braces and corsets, once routinely prescribed for chronic pain, are now used only in special cases, like spine fractures, scoliosis (a sideways curve of the spine) in teenagers or after certain surgeries. Research shows they don't offer much long-term relief. Traction, another mainstay for neck and lower back issues, has also fallen out of favour. It might ease pain temporarily but doesn't fix the underlying problem, and modern surgical techniques have largely rendered it obsolete. Even spinal injections—especially repeated epidural steroid shots—are being reconsidered due to their short-lived relief and inherent risks. Instead, newer biological therapies and more targeted interventions are slowly gaining ground.

The biggest leap in spine care has been the rise of minimally invasive surgery (MIS). Instead of cutting through large sections of muscle, surgeons can now access the spine through keyhole-sized openings, using slender tools and cameras to fix slipped discs, correct scoliosis or stabilise unstable vertebrae. Techniques like MIS-TLIF (Transforaminal Lumbar Interbody Fusion) and MIS-PLIF (Posterior Lumbar Interbody Fusion) are prime examples. In simple terms, these are ways to remove a damaged spinal disc and place a bone graft to fuse the surrounding vertebrae—all via incisions sometimes smaller than two centimetres. The costs? Rs 1.5 lakh to Rs 5 lakh.

"In the last two or three decades, technology has changed a lot," says Dr Sandeep Vaishya, executive director and head of neurosurgery at Fortis Memorial Research Institute, Gurugram. Along with advances in spine implants, he points to the growing use of intraoperative CT and MRI scans, which give real-time images during surgery, and neuro-navigation systems—digital roadmaps for the spine—that guide surgeons to place screws with pinpoint accuracy. "Many procedures can even be done under local anaesthesia," he adds, "with patients walking home the next day."

"The new minimal-access procedures have the same results as traditional surgery but involve much smaller incisions, less tissue injury, less requirement of blood and faster recovery"

DR BIPIN WALIA
Chairman, Neuro-surgery & Neurospine, Max Healthcare, Saket, Delhi

Robotics and augmented reality (AR) are now pushing the boundaries of spine surgery even further. "In the past decade, spine surgery has changed in ways once thought impossible," says Dr Hitesh Garg, head of ortho-spine surgery at Artemis Hospital, Gurugram. Robot-assisted systems, he explains, let surgeons control robotic arms that place screws and implants with millimetre-level precision, reducing the risk of nerve injury and helping patients heal faster. AR, meanwhile, lets surgeons see digital overlays of internal anatomy as they operate, almost like using a real-time X-ray visor. These cutting-edge tools don't come cheap: robotic spine surgeries cost around Rs 4–9 lakh, and procedures that use AR can cost even more. Backing all this are new kinds of implants—3D-printed titanium cages and expandable spacers—that are stronger, more body-friendly and can be custom-fitted to a patient's unique anatomy. For patients, this can mean shorter hospital

HOW TO CARE FOR YOUR SPINE

Good spine health isn't about sitting straight—it's about movement, strength and awareness. Small daily habits can prevent stiffness, support alignment and stop pain before it starts



POSTURE IN MOTION

Don't aim for a single 'perfect' position all day; keep moving. Shift between sitting, standing and walking to prevent stiffness and maintain spinal flexibility

"There's no single perfect posture. What matters is a dynamic, balanced lifestyle that keeps the spine supported, strong and mobile"

DR RAMAN KAPUR
Head of acupuncture, Sir Ganga Ram Hospital, New Delhi